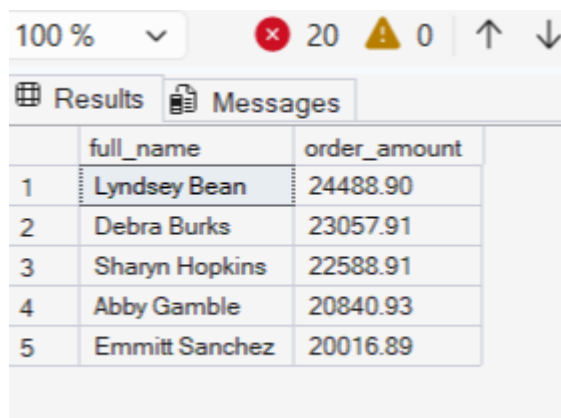


Q1. List top 5 customers by total order amount.

Retrieve the top 5 customers who have spent the most across all sales orders. Show CustomerID, CustomerName, and TotalSpent.

ANSWER : `select TOP 5`

```
    cust.first_name + ' ' + cust.last_name as full_name,  
    SUM(ordIt.list_price) as order_amount  
from sales.customers as cust  
inner join sales.orders as ord  
on ord.customer_id = cust.customer_id  
inner join sales.order_items as ordIt  
on ordIt.order_id = ord.order_id  
group by cust.first_name, cust.last_name  
order by order_amount desc
```



The screenshot shows a SQL Server query results window. The 'Results' tab is active, displaying a table with two columns: 'full_name' and 'order_amount'. The table contains five rows of data, ranked by order amount in descending order. The top row is 'Lyndsey Bean' with an order amount of 24488.90. The bottom row is 'Emmitt Sanchez' with an order amount of 20016.89. The window title bar shows '100 %' zoom, 20 rows, and 0 warnings.

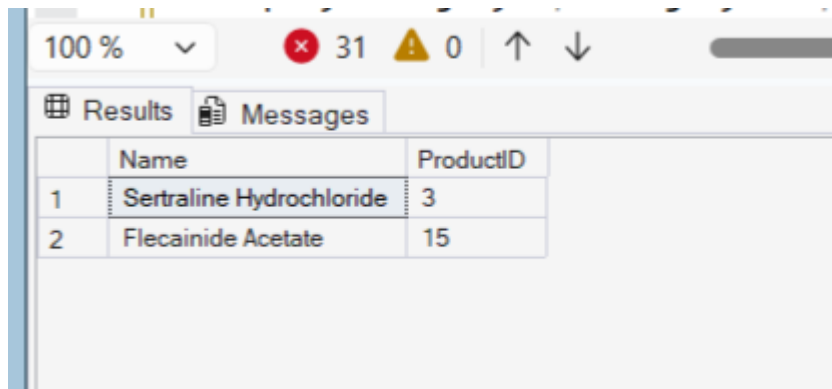
	full_name	order_amount
1	Lyndsey Bean	24488.90
2	Debra Burks	23057.91
3	Sharyn Hopkins	22588.91
4	Abby Gamble	20840.93
5	Emmitt Sanchez	20016.89

Q2. Find the number of products supplied by each supplier.

Display SupplierID, SupplierName, and ProductCount. Only include suppliers that have more than 10 products.

ANSWER : `SELECT`

```
    sup.Name,  
    COUNT(pod.ProductID) as ProductCount  
FROM  
    dbo.Supplier as sup  
INNER JOIN dbo.PurchaseOrder as po
```

	Name	ProductID
1	Sertraline Hydrochloride	3
2	Flecainide Acetate	15

Q4. For each category, find the most expensive product.

Display CategoryID, CategoryName, ProductName, and Price. Use a subquery to get the max price per category.

ANSWER : `SELECT`

`c.CategoryID,`

`c.Name as CategoryName,`

`p.Name AS ProductName,`

`p.Price`

`FROM dbo.Product p`

`INNER JOIN dbo.Category c`

`ON c.CategoryID = p.CategoryID`

`WHERE p.Price =`

`(`

`SELECT MAX(p2.Price)`

`FROM dbo.Product p2`

`WHERE p2.CategoryID = p.CategoryID`

`);`

	CategoryID	CategoryName	ProductName	Price
1	20	Hard Tile & Stone	Butalbital, Acetaminophen and Caffeine	85.95
2	19	Epoxy Flooring	Entex T	2012.38
3	18	Wall Protection	Natural Cherry Honey Herb Throat Drops	2502.47
4	17	Termite Control	SohMed Cold Relief	2813.71
5	16	Elevator	Sertraline Hydrochloride	2896.17
6	15	Hard Tile & Stone	Quetiapine Fumarate	1831.84
7	14	Glass & Glazing	Losartan Potassium	2749.20
8	13	NULL	COMETRIQ	2778.36
9	12	Asphalt Paving	Acetaminophen PM	2677.44
10	10	Exterior Signage	Flecainide Acetate	2325.87
11	9	Termite Control	Secura Moisturizing Cleanser	2627.59
12	7	Fire Protection	Oral Analgesic	2980.75
13	6	Electrical and Fire Alarm	Fluoxetine	726.23
14	5	Soft Flooring and Base	Isopropyl Alcohol Surgical Scrub	2554.94
15	4	Painting & Vinyl Wall Covering	Focalin	2455.81
16	3	Framing (Wood)	ILLUMINATING DAILY MOISTURIZER	2437.99
17	2	Landscaping & Irrigation	Betapace AF	1543.64

Q5. List all sales orders with customer name, product name, category, and supplier.

For each sales order, display:

OrderID, CustomerName, ProductName, CategoryName, SupplierName, and Quantity.

ANSWER : `select`

```

so.OrderID,
cust.Name as CustomerName,
pr.Name as ProductName,
cat.Name as CategoryName,
sup.Name as SupplierName,

```

```
sod.Quantity
```

```
from dbo.SalesOrder as so
```

```
inner join dbo.Customer as cust
```

```
on cust.CustomerID = so.CustomerID
```

```
inner join dbo.SalesOrderDetail as sod
```

```
on sod.OrderID = so.OrderID
```

```
inner join dbo.Product as pr
```

```
on pr.ProductID = sod.ProductID
```

```
inner join dbo.Category as cat
```

```
on cat.CategoryID = pr.CategoryID
```

```
inner join PurchaseOrderDetail as pod
```

```
on pod.ProductID = pr.ProductID
```

```
inner join PurchaseOrder as po
```

on po.OrderID = pod.OrderID

inner join Supplier as sup

on sup.SupplierID = po.SupplierID

Results		Messages				
	OrderID	CustomerName	ProductName	CategoryName	SupplierName	Quantity
1	21	Marcela	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	31
2	27	Sayres	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	12
3	33	Kristina	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	11
4	9	Ewie	Dexamethasone	Landscaping & Irrigation	Kassulke LLC	25
5	76	Kristina	Sertraline Hydrochloride	Elevator	Bogisch-Block	20
6	29	Oriana	SALSOLA KALI POLLEN	Fire Protection	Abshire Inc	37
7	20	Oriana	COMETRIQ	NULL	Mann, Will and Fritsch	15
8	29	Oriana	SALSOLA KALI POLLEN	Fire Protection	Bogisch-Block	37
9	18	Lawton	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	32
10	36	Lucas	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	28
11	62	Eddy	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	17
12	73	Courtney	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	40
13	87	Lynn	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	40
14	44	Xenia	SohMed Cold Relief	Termite Control	Schmidt LLC	14
15	55	Lawrence	SohMed Cold Relief	Termite Control	Schmidt LLC	11
16	70	Pattie	SohMed Cold Relief	Termite Control	Schmidt LLC	34
17	72	Huntington	SohMed Cold Relief	Termite Control	Schmidt LLC	24

Q6. Find all shipments with details of warehouse, manager, and products shipped.

Display:

ShipmentID, WarehouseName, ManagerName, ProductName, QuantityShipped, and TrackingNumber.

ANSWER : **SELECT**

s.ShipmentID,

w.Capacity **AS** WarehouseName,

e.**Name AS** ManagerName,

p.**Name AS** ProductName,

sd.Quantity **AS** QuantityShipped,

s.TrackingNumber

FROM dbo.Shipment s

INNER JOIN dbo.Warehouse w

ON w.WarehouseID = s.WarehouseID

INNER JOIN dbo.Employee e

ON e.EmployeeID = w.ManagerID

INNER JOIN dbo.ShipmentDetail sd

ON sd.ShipmentID = s.ShipmentID

INNER JOIN dbo.**Product** p

ON p.ProductID = sd.ProductID

ORDER BY s.ShipmentID;

Results Messages

	ShipmentID	WarehouseName	ManagerName	ProductName	QuantityShipped	TrackingNumber
1	1	22411	Daloris	Midol Complete	23	NULL
2	2	63723	Rose	Quetiapine Fumarate	25	NULL
3	3	22411	Daloris	Secura Moisturizing Cleanser	23	NULL
4	4	25835	Rose	Entex T	14	NULL
5	5	22411	Daloris	Promethazine Hydrochloride	29	NULL
6	6	92228	Neille	PASPALUM NOTATUM POLLEN	6	NULL
7	7	97686	Aurora	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	1	NULL
8	8	92228	Neille	Icy Hot Back and Large Areas	8	NULL
9	9	25835	Rose	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	7	NULL
10	10	25835	Rose	COMETRIQ	14	NULL
11	11	22548	Aurora	Natural Cherry Honey Herb Throat Drops	20	NULL
12	12	25835	Rose	Icy Hot Back and Large Areas	30	NULL
13	13	22411	Daloris	ILLUMINATING DAILY MOISTURIZER	18	NULL
14	14	22411	Daloris	Midol Complete	18	NULL
15	15	97686	Aurora	Maximum Strength	15	NULL
16	16	22411	Daloris	Oral Analgesic	17	NULL
17	17	92458	Rose	Entex T	22	NULL
18	18	92228	Neille	Quetiapine Fumarate	21	NULL
19	19	22411	Daloris	Dexamethasone	14	NULL
20	20	63723	Rose	Sertraline Hydrochloride	6	NULL

Query executed successfully

Q7. Find the top 3 highest-value orders per customer using RANK(). Display CustomerID, CustomerName, OrderID, and TotalAmount.

ANSWER : WITH CustomerOrderRank AS

```
(
SELECT
    c.CustomerID,
    c.Name AS CustomerName,
    so.OrderID,
    so.TotalAmount,
    RANK() OVER
    (
        PARTITION BY c.CustomerID
        ORDER BY so.TotalAmount DESC
    ) AS OrderRank
FROM dbo.Customer c
INNER JOIN dbo.SalesOrder so
    ON so.CustomerID = c.CustomerID
)
SELECT
    CustomerID,
```

CustomerName,

OrderID,

TotalAmount

FROM CustomerOrderRank

WHERE OrderRank <= 3

ORDER BY CustomerID, TotalAmount DESC;

Results Messages

	CustomerID	CustomerName	OrderID	TotalAmount	
1	2	Daune	97	29549.27	
2	3	Nevile	68	53370.84	
3	3	Nevile	93	46081.93	
4	4	Eustace	3	29705.50	
5	6	Murray	16	12849.15	
6	7	Lem	91	73192.12	
7	11	Dory	43	51043.81	
8	12	Dwight	5	91621.22	
9	12	Dwight	52	59504.03	
10	12	Dwight	6	24593.95	
11	13	Liliane	46	44724.06	
12	16	Sayres	27	22896.23	
13	19	Xenia	44	76595.26	
14	19	Xenia	74	61123.59	
15	21	Lucais	36	50207.49	
16	21	Lucais	57	44102.74	
17	24	Eddy	62	80551.84	
18	25	Lief	41	58133.77	
19	25	Lief	47	32650.50	
20	26	Talbot	66	62369.56	

Query executed successfully.

Q8. For each product, show its sales history with the previous and next sales quantities (based on order date). Display ProductID, ProductName, OrderID, OrderDate, Quantity, PrevQuantity, and NextQuantity.

ANSWER : SELECT

p.ProductID,

p.Name AS ProductName,

so.OrderID,

so.OrderDate,

sod.Quantity,

```

LAG(sod.Quantity) OVER
(
    PARTITION BY p.ProductID
    ORDER BY so.OrderDate
) AS PrevQuantity,

LEAD(sod.Quantity) OVER
(
    PARTITION BY p.ProductID
    ORDER BY so.OrderDate
) AS NextQuantity

FROM dbo.Product p
INNER JOIN dbo.SalesOrderDetail sod
    ON sod.ProductID = p.ProductID
INNER JOIN dbo.SalesOrder so
    ON so.OrderID = sod.OrderID
ORDER BY
    p.ProductID,
    so.OrderDate;

```

	ProductID	ProductName	OrderID	OrderDate	Quantity	PrevQuantity	NextQuantity
1	1	Butalbital, Acetaminophen and Caffeine	79	2003-04-21 04:11:35	19	NULL	17
2	1	Butalbital, Acetaminophen and Caffeine	61	2016-05-18 03:37:40	17	19	13
3	1	Butalbital, Acetaminophen and Caffeine	88	2017-07-11 14:02:27	13	17	NULL
4	2	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	30	2000-12-26 15:50:29	23	NULL	NULL
5	3	Sertraline Hydrochloride	76	2009-10-24 02:52:28	20	NULL	NULL
6	4	SohMed Cold Relief	72	2003-11-14 11:53:28	24	NULL	14
7	4	SohMed Cold Relief	44	2004-04-04 09:26:01	14	24	11
8	4	SohMed Cold Relief	55	2012-04-24 07:46:10	11	14	34
9	4	SohMed Cold Relief	70	2021-08-30 09:18:44	34	11	NULL
10	5	Promethazine Hydrochloride	15	2000-05-29 09:37:55	13	NULL	23
11	5	Promethazine Hydrochloride	80	2017-08-05 16:48:19	23	13	NULL
12	6	Ondansetron	23	2002-02-06 08:48:16	14	NULL	24
13	6	Ondansetron	86	2011-05-18 11:07:42	24	14	37
14	6	Ondansetron	53	2013-01-29 02:06:34	37	24	35
15	6	Ondansetron	50	2020-10-08 05:33:26	35	37	NULL
16	7	TopCare Antacid Calcium	43	2007-12-19 07:08:49	15	NULL	29
17	7	TopCare Antacid Calcium	94	2008-03-01 03:58:21	29	15	NULL
18	8	Maximum Strength	17	2015-07-29 20:13:43	21	NULL	NULL
19	9	Secura Moisturizing Cleanser	56	2003-01-16 13:11:31	34	NULL	18
20	9	Secura Moisturizing Cleanser	28	2014-10-07 17:32:02	18	34	22

Q9. Create a view named vw_CustomerOrderSummary that shows for each customer:

CustomerID, CustomerName, TotalOrders, TotalAmountSpent, and LastOrderDate.

ANSWER : CREATE VIEW vw_CustomerOrderSummary

AS

SELECT

c.CustomerID,

c.Name AS CustomerName,

COUNT(so.OrderID) AS TotalOrders,

SUM(so.TotalAmount) AS TotalAmountSpent,

MAX(so.OrderDate) AS LastOrderDate

FROM dbo.Customer c

LEFT JOIN dbo.SalesOrder so

ON so.CustomerID = c.CustomerID

GROUP BY

c.CustomerID,

c.Name;

GO

SELECT *

FROM vw_CustomerOrderSummary;

100 %

37

Results

Messages

	CustomerID	CustomerName	TotalOrders	TotalAmountSpent	LastOrderDate
1	1	Pavla	0	NULL	NULL
2	2	Daune	1	29549.27	2008-12-11 11:36:26
3	3	Nevile	2	99452.77	2010-11-30 13:33:08
4	4	Eustace	1	29705.50	2019-12-22 19:57:18
5	5	Elga	0	NULL	NULL
6	6	Murray	1	12849.15	2008-07-06 01:10:56
7	7	Lem	1	73192.12	2012-03-18 16:38:10
8	8	Reube	0	NULL	NULL
9	9	Sergeant	0	NULL	NULL
10	10	Valeria	0	NULL	NULL
11	11	Dory	1	51043.81	2007-12-19 07:08:49
12	12	Dwight	3	175719.20	2020-07-21 05:01:11
13	13	Liliane	1	44724.06	2012-12-21 02:35:15
14	14	Juditha	0	NULL	NULL
15	15	Claudianus	0	NULL	NULL
16	16	Sayres	1	22896.23	2022-11-16 00:14:29
17	17	Leslie	0	NULL	NULL
18	18	Waring	0	NULL	NULL
19	19	Xenia	2	137718.85	2007-06-18 06:30:46
20	20	Jermaine	0	NULL	NULL

Q10. Write a stored procedure sp_GetSupplierSales that takes a SupplierID as input and returns the total sales amount for all products supplied by that supplier.

ANSWER : **CREATE PROCEDURE** sp_GetSupplierSales

@SupplierID **INT** = NULL

AS

BEGIN

SET NOCOUNT ON;

SELECT

s.SupplierID,

s.Name **AS** SupplierName,

ISNULL(**SUM**(sod.TotalAmount),0) **AS** TotalSalesAmount

FROM dbo.Supplier s

INNER JOIN

(

SELECT DISTINCT

po.SupplierID,

```

        pod.ProductID

FROM dbo.PurchaseOrder po

INNER JOIN dbo.PurchaseOrderDetail pod

    ON pod.OrderID = po.OrderID

) sp

    ON sp.SupplierID = s.SupplierID

INNER JOIN dbo.SalesOrderDetail sod

    ON sod.ProductID = sp.ProductID

WHERE (@SupplierID IS NULL OR s.SupplierID = @SupplierID)

GROUP BY

    s.SupplierID,

    s.Name;

END;

EXECUTE sp_GetSupplierSales;

```

Results		Messages	
	SupplierID	SupplierName	TotalSalesAmount
1	9	Sawayn-Adams	187476.80
2	3	Abshire Inc	193075.62
3	6	Abernathy-Waelchi	176188.25
4	7	Kassulke LLC	206335.95
5	1	Schamberger Inc	187958.76
6	10	Stokes, Funk and Herman	226584.29
7	4	Schmidt LLC	197149.70
8	5	Mann, Will and Fritsch	174312.48
9	2	Bogisich-Block	194856.38
10	8	Lang-Rodriguez	211594.57